WRITING SAMPLE · SPATIAL ANALYSIS

Mapping Accident Risk for Cyclists and Pedestrians in Marion County

A geographic information systems analysis of speed-limit risk on shared bikeways, bikeway–pedestrian conflict points and proposed bike share station locations in Indianapolis.

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ACADEMIC WORK NOTICE

This report was originally prepared as graduate coursework at Indiana University's O'Neill School of Public and Environmental Affairs and has been reformatted for portfolio presentation.

The research design, spatial analysis and maps are the author's original work, built from public data published by the Indiana Department of Transportation, the City of Indianapolis (Indy Open Data Portal and IndyGIS) and the U.S. Census Bureau. Prepared for portfolio and professional evaluation purposes.

Introduction

Walking and cycling are growing as everyday transportation in American cities, and planning safe walk-bike infrastructure depends on knowing where people on foot and on bikes are most exposed to harm.

Active transportation plays a key role in planning sustainable urban infrastructure. As walking and bicycling become more common forms of utilitarian travel, traffic safety must be part of sustainable transportation planning. Infrastructure design has clear safety implications, especially at intersections and where motor vehicles travel faster than 25 miles per hour.

Studies suggest that high proportions of commuter cyclists are associated with lower risks of serious injury or death, a "safety in numbers" effect (Vandenbulcke-Plasschaert, 2011). Building bicycle-only tracks and marking bicycle lanes increases bicycle traffic and decreases motor vehicle traffic (Jensen, 2008). Planners therefore need countermeasures that accommodate growing urban cycling, along with safe locations for new bike share stations (Wang et al., 2016).

Research questions

1. What risks do speed limits pose to cyclists on on-street and sharrow bikeways?

Knowing where cyclists are at risk is the starting point for new safety measures. Identifying high-risk bikeways allows a spatial analysis of the relationship between high speed limits and cyclist safety. When a cyclist collides with a motor vehicle, the cyclist is far more likely than the driver to be severely injured (Wegman et al., 2012).

2. Where do bikeways and pedestrian networks intersect, and what risk does that create?

Just as cyclists are more likely to be seriously injured in a collision with a motor vehicle, pedestrians are more likely to be seriously injured in a collision with a cyclist (Larsen et al., 2013). Proximity analysis gives a closer look at this risk relationship. Population density is displayed alongside it as a visual aid for estimating the likelihood of pedestrian–cyclist conflicts at each intersection.

3. Where should new bike share stations go?

Strategically placed bike share stations are associated with greater exposure to, and use of, those facilities (Hasani et al., 2019). Using density analysis, potential station locations are identified in relation to the density of pedestrian–bikeway intersections. Multi-criteria decision analysis is then used to recommend station sites that would increase public use and strengthen the "safety in numbers" effect.

Data sources

Indiana Department of Transportation

INDOT provided street centerline data and 2019 annual average daily traffic reports (Indiana Department of Transportation, n.d.). Street centerline attributes allowed speed limits to be mapped across Marion County. Traffic volumes were not displayed in this analysis but would strengthen future work on safety countermeasures. Bicycle crash locations would be the most valuable addition to future research on risk; however, INDOT publishes mostly aggregate motor vehicle traffic and crash data.

Indy Open Data Portal

The City of Indianapolis open data portal supplied pedestrian network line data and the locations of existing bike share stations (Indy Open Data Portal, n.d.). The data has labeling weaknesses: only one multiuse path is coded "off-road" rather than "off-street," and several bikeways and multiuse paths are marked "unclassified" without explanation. To preserve its integrity, the data was used as published, without assumptions. Future research would benefit from corrected labels.

IndyGIS

IndyGIS provided the most useful information on bicycle traffic: line data for bikeways and greenways, separated by type, and the Marion County boundary (IndyGIS, n.d.). This data was well suited to representing the different types of bikeway and to setting the boundary of the analysis.

U.S. Census Bureau

2010 Census block group data was the most recent available for calculating and mapping population density (United States Census Bureau, n.d.). Block groups give a more precise view of population at key locations than larger census geographies.

Analysis

Spatial analysis: speed limits on shared bikeways

Mapping speed limits against bikeways and sharrow routes reveals where cyclists face the greatest threat. Collisions with motor vehicles are most likely on on-street bikeways and sharrow routes, where bicycle and motor vehicle traffic share the road (Figure 1). Several long stretches of on-street bikeway carry a 50 mph speed limit, most notably on Lafayette Road between West 62nd Street and West 30th Street.

These findings give decision-makers a basis for action: building protected bikeways on high-risk corridors or, on shorter high-risk segments, lowering the speed limit. Either would reduce cyclist risk, with priority guided by population density and crash history.

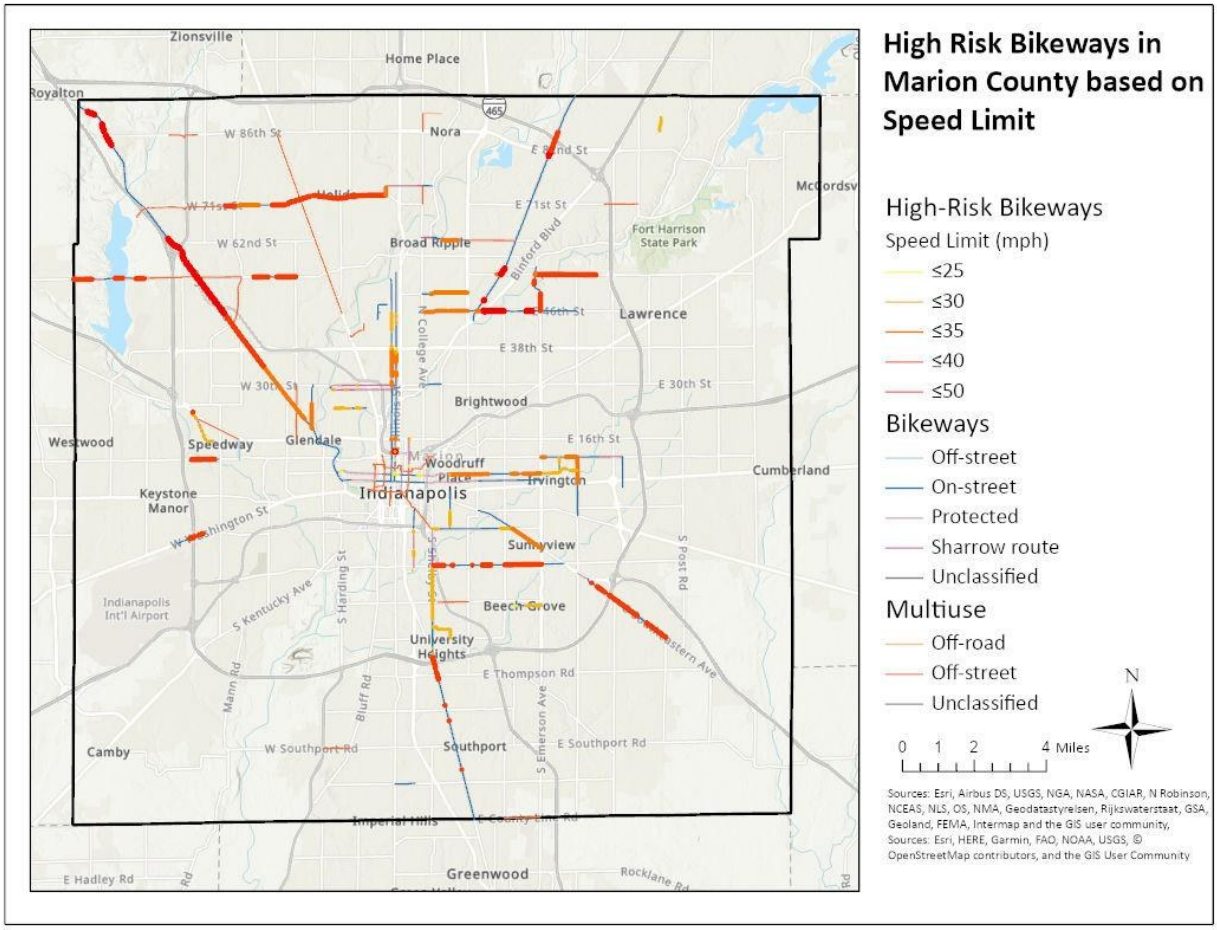


Figure 1. High-risk bikeways in Marion County based on posted speed limit, by bikeway type.

Proximity analysis: bikeway–pedestrian conflict points

Identifying where bikeways cross pedestrian networks highlights where cyclist–pedestrian collisions are most likely (Figure 2). Knowing where these intersections are, and how far they sit from main bicycle routes, helps cyclists plan routes that avoid them.

Bikeway–pedestrian intersections cluster on the west side of downtown Indianapolis, where pedestrian paths cross protected bikeways. In Speedway, off-street bikeways and pedestrian networks intersect closely. Pedestrians are not permitted on on-street bikeways, so in urban areas the two networks rarely meet there. Residential street intersections are the exception, as both cyclists and pedestrians use them. The south side of Indianapolis, with many on-street bikeway–pedestrian intersections, illustrates this pattern.

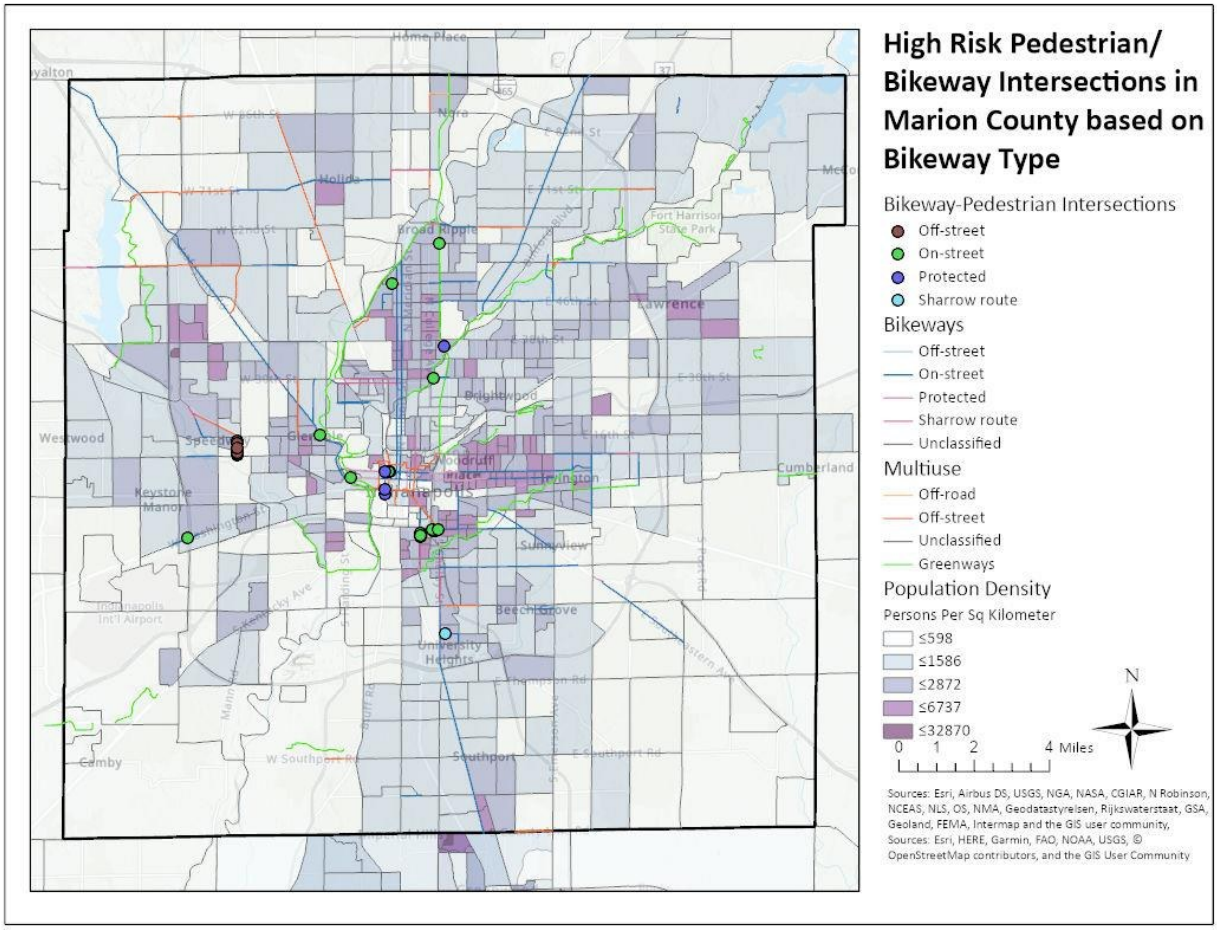


Figure 2. High-risk pedestrian–bikeway intersections in Marion County by bikeway type, shown over population density by census block group.

Density analysis: proposed bike share stations

Kernel density estimation identifies "hot spots" of pedestrian activity along bikeways, which is central to understanding where bike share stations would be used (Figure 3). As Table 1 shows, on-street, protected and sharrow bikeways do not allow pedestrians, while off-street bikeways, multiuse paths and greenways allow both.

Facility	Type	Cyclists allowed	Pedestrians allowed
Bikeway	Off-street	Yes	Yes
Bikeway	On-street	Yes	No
Bikeway	Protected	Yes	No
Bikeway	Sharrow route	Yes	No
Bikeway	Unclassified	Yes	No
Multiuse	Off-road	Yes	Yes
Multiuse	Off-street	Yes	Yes
Multiuse	Greenway	Yes	Yes
Multiuse	Unclassified	Yes	Yes

Table 1. Permitted users by bikeway and multiuse facility type, Marion County.

Bikeway–pedestrian intersections are most concentrated on the near west side, around the IUPUI campus, where two new stations are proposed. A second high-density area in the northwest quadrant of downtown warrants another station. At the southern edge of the current network, west of Beech Grove along Meridian Street, an additional station would encourage cycling on one of the main routes between the south side and downtown.

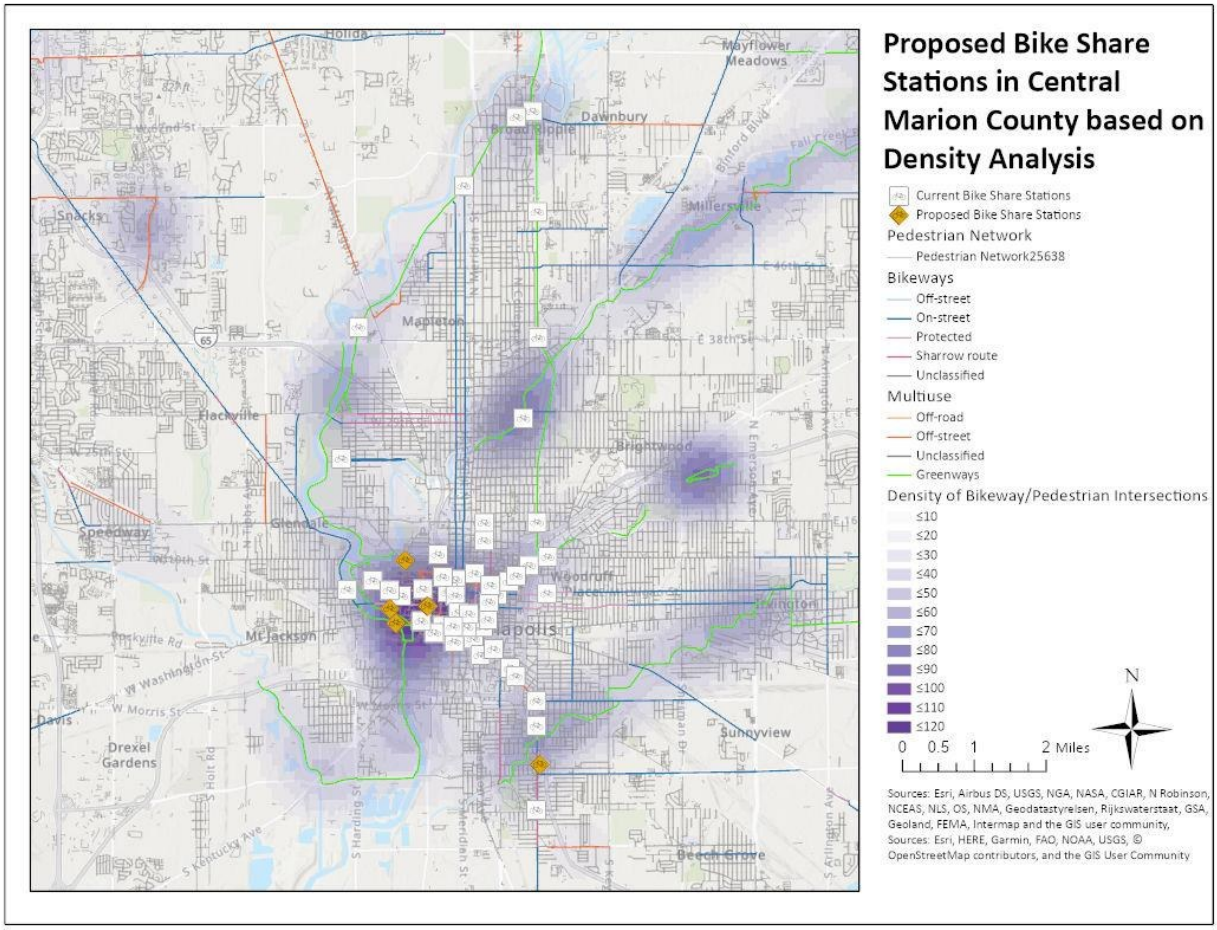


Figure 3. Proposed bike share stations in central Marion County based on kernel density of pedestrian–bikeway intersections.

Conclusion

Because transportation systems are spatial, geographic information systems are essential to comprehensive bicycle safety planning. Safe active transportation is central to sustainable urban planning goals. The "safety in numbers" effect, in which safety improves as more people cycle, justifies the proposal of five additional bike share stations in pedestrian areas with high exposure to bikeways, as measured by the density of pedestrian–bikeway intersections.

Improving bicycle safety and use calls for multi-faceted strategies built on empirically evaluated countermeasures. Future research could add crash locations, traffic volumes and elevation data; elevation was left out here because Marion County's terrain is relatively flat. Route-optimization tools within GIS could also help bicycle courier services plan the safest and fastest routes. With accurate data, GIS offers planners a powerful set of visual tools for designing active transportation routes that encourage use, safety, health and sustainable living.

References

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